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Paleomagnetic results from the Upper Carboniferous of the Shan–Thai–Malay block of western Yunnan, China

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ABSTRACT

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The Upper Carboniferous basaltic Woniusi Formation was sampled at 21 sites from two limbs of a syncline near Baoshan city (25.2° N, 99.3° E), and from a monocline near Yongde county (23.9° N, 99.2° E) for paleomagnetic study. The study area is generally thought to be part of the Shan–Thai–Malay microplate. Thermal demagnetization reveals a single-polarity characteristic remanent magnetization (ChRM) which is carried by both magnetite and hematite, with the former being the dominant mineral in basalt and the latter being the principal carrier in associated tuffaceous basalt and sandstone. The tilt-corrected formation mean from Baoshan ($D = 209.2°$, $I = 61.0$, $\alpha_{95} = 6.7°$) confirms the results reported by previous workers, and a positive fold test constrains the ChRM acquisition to be prefolding. The paleomagnetic inclinations determined from this study are similar to that derived from the Devonian rocks from the same area by other workers, indicating a paleolatitude of about 42° for the Baoshan area during the Devono-Carboniferous time. The paleomagnetic declinations, however, present a large difference between the two sampling localities for the Upper Carboniferous as well as between the sampling sites for the Devonian and for the Upper Carboniferous. The paleo-declination difference can be partly explained by the apparent polar wander of Gondwana if the Shan–Thai–Malay microplate was rigidly attached to Gondwanaland during this time interval; the large part of it has to be ascribed to tectonic rotations between the sampling sites. This fact suggests that extreme caution should be exerted when trying to establish an apparent polar wander path for, and to reconstruct the paleo-geographic position of, the Shan–Thai–Malay microplate.

Introduction

It is now known that Asia is a collage of a number of tectonic blocks, though the origin and evolution of these continental fragments are not yet quite clear. In unraveling their derivation, timing of docking, as well as post-docking motion, paleomagnetic studies have proved to be a powerful tool. With the active involvement of western scientists, the last decade saw a rapid increase in paleomagnetic data from East Asia, especially China. One of the current trends in the paleomagnetic investigation of China is that some smaller blocks, or terranes, have begun to be explored (Zhang and Zhang, 1986; Lin and Watts, 1988; Fang et al., 1989; Chen et al., 1991) while such major cratons as the North China, South China, and Tarim blocks are being studied.

One of the terranes or continental fragments which make up Southeast Asia is the Shan–Thai–Malay block; it can be studied in the Baoshan area of western Yunnan province in southwest China. The region is generally thought to be the northern extension of the Shan–Thai–Malay microplate which composes parts of Burma, Thailand and Malaysia (Ren et al., 1980; Mitchell, 1981; Stauffer, 1983) (Fig. 1). In recent years, an increasing number of authers favor the idea that the Shan–Thai–Malay microplate was derived from Gondwanaland sometime during the Early Permian (Metcalfe, 1988) or the Early Triassic (Şengör, 1987). Zhang and zhang (1986) first re-

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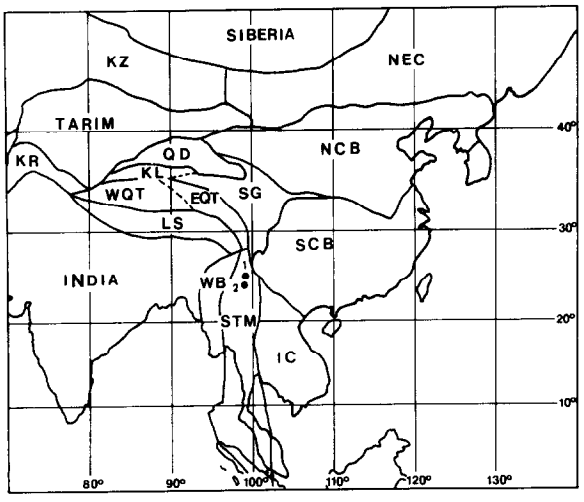


Fig. 1. The tectonic blocks of China and its environs (modified from Lin and Watts, 1988). *EQT* = East Qiangtang; *IC* = Indochina; *KL* = Kunlun; *KR* = Karakoram; *KZ* = Kazakstan; *LS* = Lhasa; *NCB* = North China; *NEC* = Northeast China; *QD* = Qaidam; *SCB* = South China; *STM* = Shan–Thai–Malay; *WB* = West Burma; *WQT* = West Qiangtang; *SG* = Songpan–Ganzi accretionary complexes. Numbered dots indicate two sampling localities in this study.

ported paleomagnetic results from the Upper Carboniferous basaltic Woniusi Formation from the Baoshan area and obtained a late Carboniferous paleolatitude of 34°S for the sampling site. Unfortunately, their results were based upon a limited number of hand samples, and no fold test was attempted, so the time of the acquisition of the magnetization was not well constrained. Shortly afterwards, Fang et al. (1989) published paleomagnetic data derived from the Devonian limestones from the same area and gave a similar paleolatitude but quite different a paleomagnetic declination. For this reason, we resampled the Woniusi Fm. at two localities in the hope of resolving this discrepacy.

Geology and sampling

Within the Chinese territory, the Shan–Thai–Malay block is bounded to the west by the Nujiang suture and to the east by the Changning-Shuangjiang suture. The exposed oldest rocks are slightly metamorphosed Lower to Middle Cambrian turbidites. Overlying them are Upper Cambrian through Jurassic platform carbonates and clastics. Volcanics occur in the Upper Carboniferous, Upper Triassic, and sparsely in the Middle Jurassic and Cenozoic.

The Upper Carboniferous Woniusi Fm. has a wide areal distribution in the Baoshan area, and probably correlates with the volcanic rocks in the Carboniferous Mergui Group in the Shan States of Burma (see Mitchell, 1981). It consists chiefly of basalt with intercalated tuffaceous sandstone, siltstone, shale, and in some places, limestone. Such fusulinids as *Triticites* and *Schwagerina*, brachiopods as *Dictyoclostus*, *Plicochonetes*, and *Chonetes*, and bryozoans as *Fistulipora* and *Fenestella* have been found in these intercalations, indicating a Late Carboniferous age for the formation and a submarine environment for the basaltic

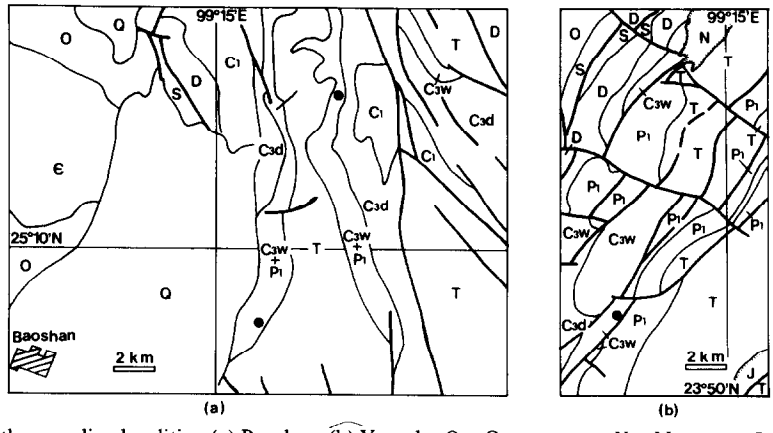


Fig. 2. Geologic maps of the sampling localities. (a) Baoshan, (b) Yongde. *Q* = Quarternary; *N* = Neogene; *J* = Jurassic; *T* = Triassic; P_l = Lower Permian; C_3w = Upper Carboniferous Woniusi Fm.; C_3d = Upper Carboniferous Dingjiazhai Fm.; C_l = Lower Carboniferous; *D* = Devonian; *S* = Silurian; *O* = Ordovician; Є = Cambrian. Black circles indicate the sampling sections.

eruption. Regionally, the thickness of the Woniusi Fm. ranges from 355 to 768 m, and two eruptive cycles, separated by sedimentary and/or volcaniclastic rocks, are generally recognizable. Conformably underlying this formation is the Upper Carboniferous Dingjiazhai Fm., which is composed primarily of pebbly sandstones and shales and contains abundant *Triticites* fusulinids and other late Carboniferous brachiopods and pelecypods. Some Chinese geologists have suggested that the pebbly sandstones and shales of the Dingjiazhai Fm. were of glacio-marine origin and correlative with the diamictites of comparable age elsewhere on the Shan–Thai–Malay block and

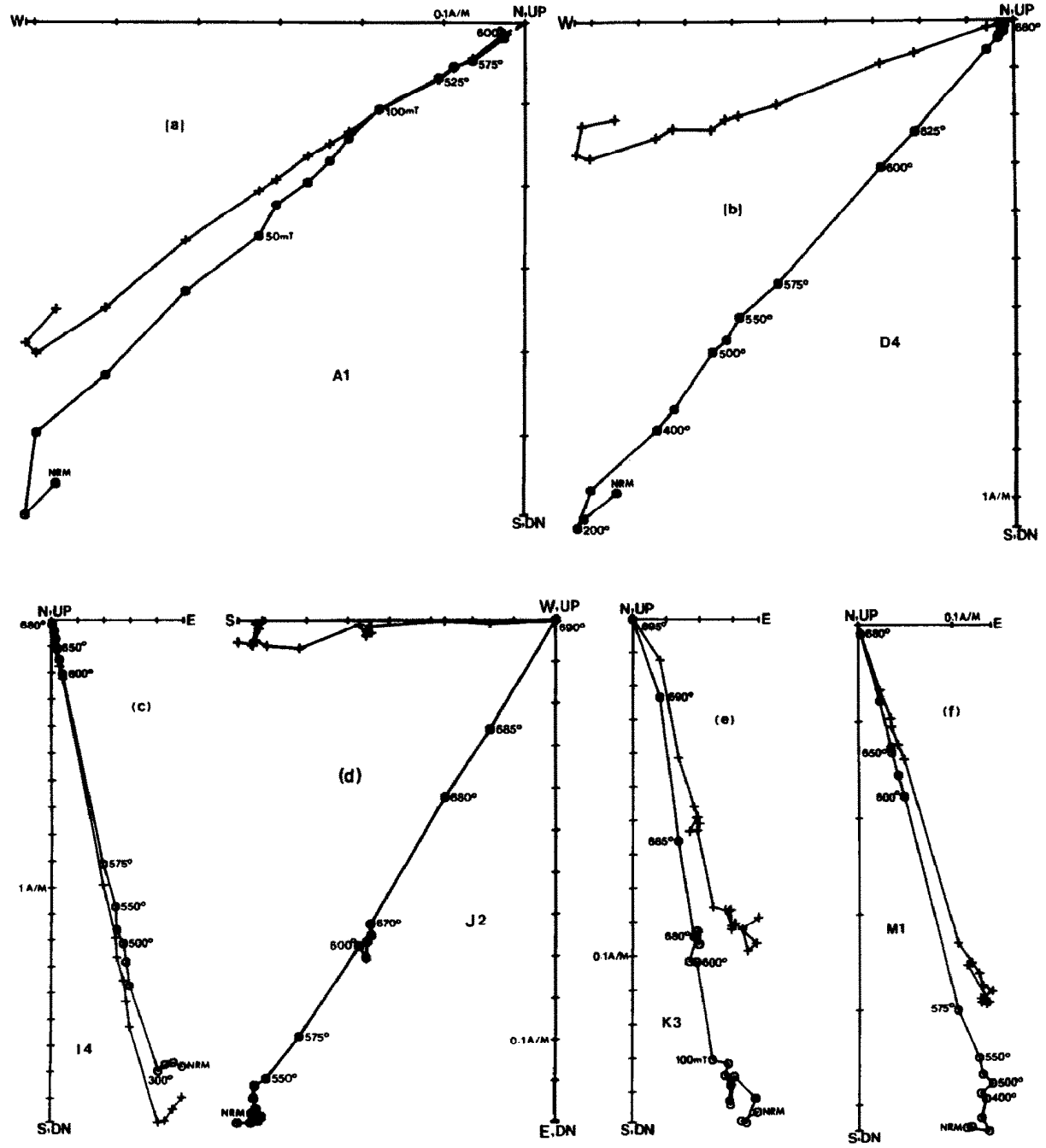


Fig. 3. Representative orthogonal plots in in situ coordinates for samples from Baoshan. Crosses (circles) plotted on horizontal (vertical) planes. Thermal treatment levels in degrees Celsius.

on the Lhasa block in Tibet (Wang, 1983; Cao, 1986). Unconformably overlying the Woniusi Fm. are the ferruginous shales of the Bingma Fm. and limestones and dolomites of the Dawazi/Yongde Fm., both yielding rich Early Permian fusulinids, brachiopods, and corals. Thus, the Woniusi Fm. is also bracketed as late Carboniferous in age by underlying and overlying formations.

The first sampling locality (25.2° N, 99.3° E) is located 12 km northeast of Baoshan city, where the lower Permian and Upper Carboniferous crop out on the two limbs of a N–S-trending, open syncline (Fig. 2a). The west limb of the syncline dips east-southeast at 17 to 35° while the east limb dips west-northwest at 17–20°. At the type section on the west limb, where Zhang and Zhang's hand samples were originally taken, the upper part of the basaltic sequence is exposed along a gulley; the outcrop is fresh and the bedding, indicated by alternations of compact basalt and amygloidal basalt, is easy to determine. Thus eight sites (A–H) were drilled in the upper part of the sequence. In order to apply a fold test, another five sites (I–M) were drilled in the comparable stratigraphic level on the east limb about 10 km north of the type section.

The second sampling locality (23.9° N, 99.2° E) is 10 km south of Yongde county, where 8 sites (N–U) were drilled at a SE-dipping monocline. The bedding attitude was determined from the tuffaceous sandstone at the top of the sequence. It was noticed that at this locality both cross and strike faults are relatively well developed (Fig. 2b) and the basalts are slightly altered.

Five cores were usually taken from each site. In all, 66 cores from Baoshan and 41 cores from Yongde were collected. A portable rock drill was used for sample extraction and an orientation device mounted with a Brunton compass was used for core orientation. The effect of the magnetization of the basaltic rocks on the compass was checked, when measuring the bedding attitude and orienting the cores, by gradually reducing the distance from the compass to the rock surface, and no significant deflection was detected.

Experiments

Laboratory work was conducted at the University of Florida. The 2.5 cm diameter cores were sliced into 2.5 cm long specimens. Both stepwise alternating field (AF) and thermal demagnetization were carried out on the pilot specimens. Increments of 10 mT in the peak field were used during AF demagnetization up to 100 mT, the maximum field of the GSD-1 Schonstedt AF demagnetizer. Thermal demagnetization was performed in a TSD-1 Schonstedt thermal demagnetizer, starting with increments of 100° below 400° C, then 50° between 400° and 500° C, 25° in the range 500–650° C, finally 10° increments to the destruction of the remanent magnetization. The pilot experiments showed that thermal demagnetization is more effective than AF tech-

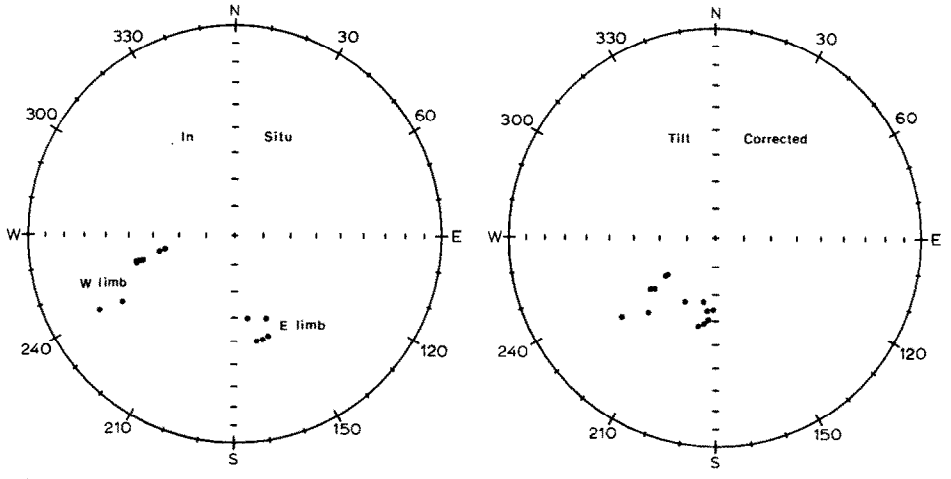


Fig. 4. Equal-area projections of the site mean directions from Baoshan, all points on the lower hemispheres.

niques in reducing the intensity of the natural remanent magnetization (NRM) of the sample (see Figs. 3a, 3e and 5a). Therefore, only thermal demagnetization was applied to the remaining samples with abbreviated steps. Remanence of all samples was measured in a DSM-2 Schonstedt spinner magnetometer, except for two samples whose intensities became so weak after heating to 500 °C that the measurement had to be shifted to an SCT two-axis cryogenic magnetometer.

Orthogonal vector plots were constructed for each specimen to examine its demagnetization behavior (Zijderveld, 1967). Principal component analysis (Kirschvink, 1980) was employed to calculate the directions represented by the linear segments in the plots. In the case of characteristic components, the directions were calculated both with and without anchoring datum points to the origin. Site and formation means were computed using Fisher (1953) statistics. Only those site means based on calculation without origin-anchoring are listed in the next section.

TABLE 1

Site and formation means statistics *

Site	n/N	Bedding	In situ		Tilt corrected				Pole position	
			$D(°)$	$I(°)$	$D(°)$	$I(°)$	α_{95}	k_2	Lat. °N	Long. °E
Baoshan (25.2°N, 99.3°E)										
A	5/5	14/21	239.4	38.4	222.4	51.2	6.7	130.47	−20.1	61.7
B	5/5	14/21	241.1	27.6	230.2	41.7	6.1	156.32	−20.9	50.6
C	5/5	14/21	255.4	50.7	232.6	67.2	2.0	1397.86	−1.7	68.5
D	5/5	14/21	255.3	50.4	232.6	66.9	4.9	241.43	−1.9	68.3
E	5/5	42/17	255.1	52.8	231.3	59.1	2.6	886.84	−9.3	61.9
F	5/5	42/17	255.2	51.9	232.1	58.3	8.9	74.81	−9.5	60.9
G	5/5	40/35	257.7	60.6	186.7	62.3	8.6	80.99	−21.0	94.1
H	5/5	40/35	258.5	63.0	181.8	63.1	4.7	266.52	−20.3	98.0
I	5/5	203/20	167.8	47.6	191.4	55.7	4.9	240.57	−27.6	88.9
J [a]	4/5	203/20	170.6	57.9	205.9	63.3	4.1	503.64	−16.1	80.5
K	6/6	203/20	158.7	55.8	191.0	65.7	2.3	824.54	−16.2	91.7
L	5/5	203/20	164.5	47.9	188.2	57.2	6.9	123.96	−26.5	92.0
M	5/5	203/20	161.1	48.7	185.3	59.0	2.8	751.01	−24.9	94.8
Formation mean:										
	13/13		220.5	58.3	209.2	61.0	6.7		−17.2	77.5
			$k_1 = 8.63$,		$k_2 = 39.82$				$A_{95} = 9.3$	
Yongde (23.9°N, 99.2°E)										
N	5/5	22/37	286.1	23.0	281.1	59.7	3.1	618.60	23.4	44.9
O	5/5	22/37	283.0	19.5	276.7	55.7	3.0	635.14	19.0	41.3
P	5/5	22/37	306.7	27.7	320.5	62.5	5.7	181.79	52.1	50.8
Q	5/5	22/37	301.2	27.4	310.6	63.5	4.6	275.69	45.0	49.8
R	5/5	22/37	290.7	33.4	288.9	70.4	4.5	294.16	30.1	59.7
S	5/5	22/37	292.9	29.2	293.9	66.2	2.3	1147.00	33.3	52.8
T	6/6	10/40	289.5	56.3	61.7	81.5	13.6	25.25	30.9	116.3
U	5/5	10/40	313.9	71.2	75.9	63.7	14.6	28.27	26.4	148.8
Formation mean (without sites T and U):										
	6/8		293.3	26.9	294.5	63.9	7.8		33.9	49.6
			$k_1 = 73.72$,		$k_2 = 74.24$				$A_{95} = 11.6$	

* n/N = number of samples or sites in calculation for the means/number of samples or sites measured; bedding = strike/dip; D, I = declination and inclination; α_{95}, A_{95} = radius of the circle of 95% confidence about the mean direction and pole, respectively; k_1, k_2 = Fisher precision parameters before and after tilt correction.

[a] One sample was misoriented.

Results

Baoshan

The NRM intensities of the samples range from 0.01 to 6 A/m. A soft component, which is in most cases northerly and up in geographic coordinates, was often removed below 10 mT by AF demagnetization or below 200 °C during thermal cleaning (Fig. 3). With incremental heating, two major types of demagnetization behavior can be recognized in terms of unblocking temperature spectra. In the first type of behavior, a ChRM component was usually isolated in the range 300–600 °C and discretely unblocked between 550–600 °C. Clearly, magnetite is the magnetic carrier (Figs. 3a and 3c). In some samples, however, the ChRM component persists up to 670 °C, implying some contribution from hematite (Fig. 3c). Samples from sites A, B, C,F, G, H, and I fall into this

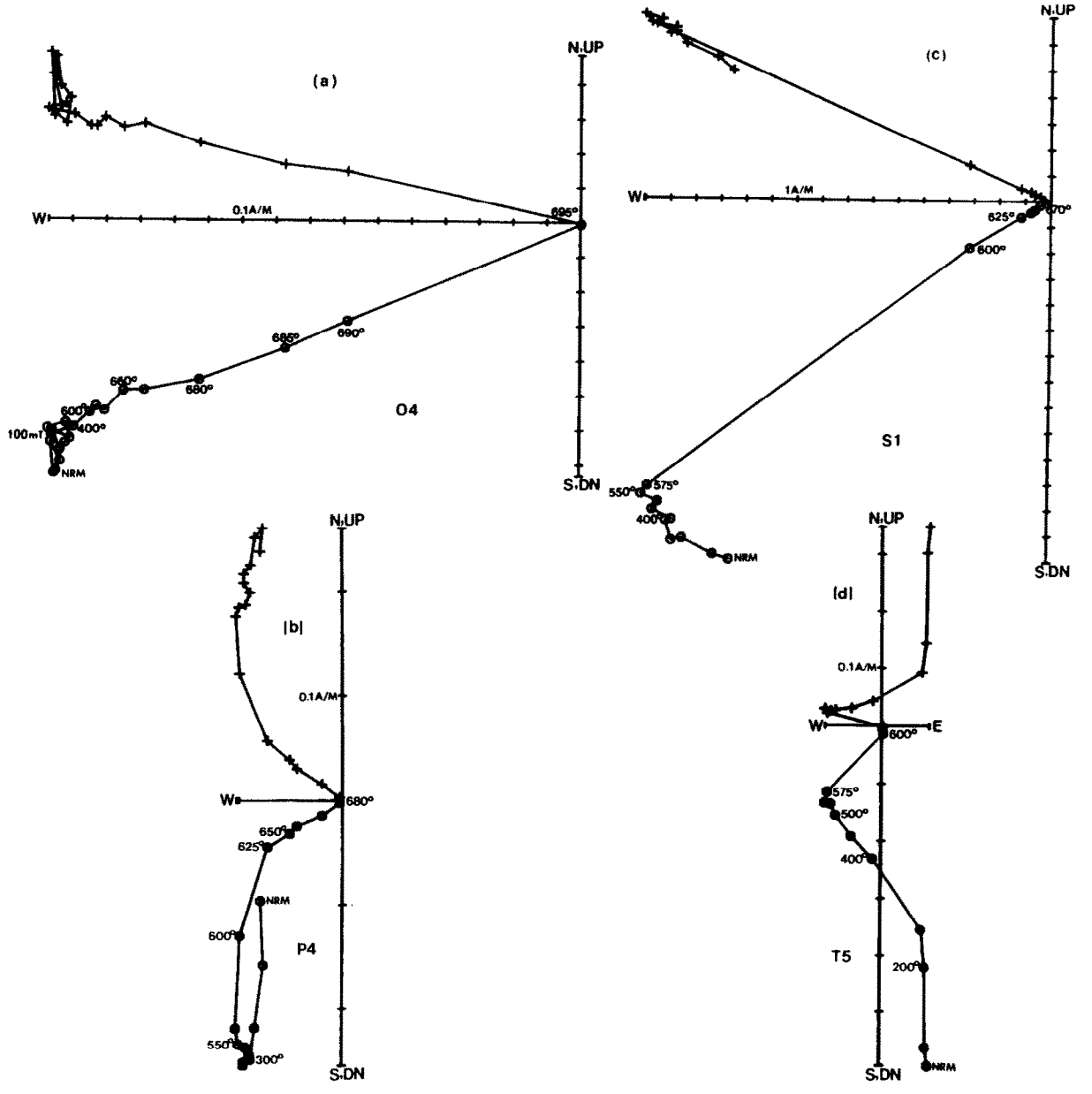


Fig. 5. Representative orthogonal plots in in situ coordinates for samples from Yongde. Symbols as in Fig. 3.

category. A second type of behavior is shown by the tuffaceous basalt samples from sites J, K, and L. The characteristic component is of the same direction as that in the samples of the first type behavior. It was isolated above 550°C and dominates the NRM in the temperature range 660–690°C, indicating that hematite is the major carrier (Figs. 3d and 3e). The red color of the samples from sites J and K also points to the presence of hematite. Samples from sites D, F, and M show big drops in NRM intensity in the range 550–600°C and above 625°C (Figs. 3b and 3f), reflecting comparable contributions from magnetite and hematite. The ChRM directions in all cases are west-southwesterly or south-southeasterly and moderately down in geographic coordinates, and are certainly not of present or recent origin.

Another indication for the early acquisition of the ChRM direction comes from the fold test. As can be seen from Table 1 and Fig. 4, the in-situ site mean directions from the two limbs of the syncline are well away from one another; the tilt-corrected directions, however, merge into a well defined group, resulting in a sharp increase of precision parameter (k) from 8.63 to 39.82. The fold test is positive at the 99% confidence level (McElhinny, 1964), so the ChRM is prefolding. Only one polarity is present, which is expected in Upper Carboniferous rocks. The thirteen tilt-corrected site means together yield a formation mean direction of 209.2°/61.0° with $\alpha_{95} = 6.7°$.

Yongde

The NRM intensities of the samples from Yongde are in the range 0.17–3.8 A/m, not as variable as those from Baoshan. The demagnetization behaviors of the samples closely corresponds to those from Baoshan. For instance, AF demagnetization hardly affected the tuffaceous sandstone samples from sites N and O (Fig. 5a), whereas thermal cleaning effectively resolved a discrete, characteristic component between 600°C and 695°C, much like the red tuffaceous basalt samples from Baoshan in which hematite is presumably the principal magnetic carrier. On the other hand, the response of the basalt samples from sites Q, R, and S to demagnetization is very similar to the first type behavior described earlier (compare Fig. 5c with Fig. 3c), in which magnetite is the dominant magnetic mineral. Some samples did show more complex demagnetization behavior, which can be exemplified by samples from site P (Fig. 5b). Samples from this site possess three components with overlapping blocking temperature spectra; the ChRM, which is northwesterly and steeply down in stratigraphic coordinates, similar to those isolated from sites N, O, Q, R, and S, was not resolved until above 625°C. Unfortunately, this is not the case for sites T and U. Most samples from these two sites showed multivectorial magnetization (Fig. 5d) and often abruptly unblocked between 575°C and 600°C, resulting in a very steep direction and much larger

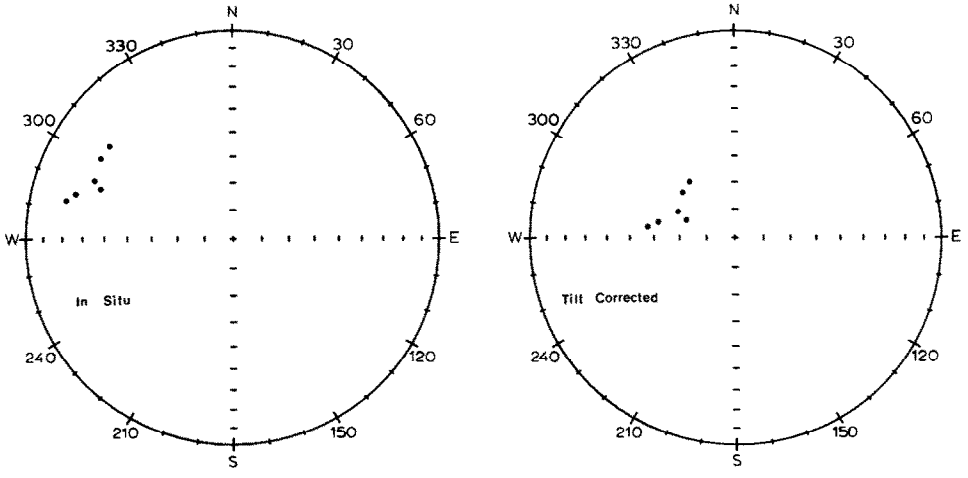


Fig. 6. Equal-area projections of the site mean directions from Yongde, all points on the lower hemispheres.

α_{95} for the site mean directions (see Table 1). Therefore, the formation mean for this locality (294.5°/63.9°, $\alpha_{95} = 7.8°$) was calculated excluding the last two sites. The equal-area projection of the site mean directions for Yongde is shown in Fig. 6. Again, only one polarity was observed.

Discussion

The tilt-corrected mean direction for the Woniusi Fm. at Baoshan is similar to that (218.5°/53.6°, $\alpha_{95} = 5.9°$) reported by Zhang and Zhang (1986); the positive fold test achieved in this study further constrains this paleomagnetic direction to be prefolding. Since the folding in the study area probably took place in the late Mesozoic (Yanshanian) or early Cenozoic (Himalayan) (Zhong Dalai, pers. commun., 1990), the prefolding magnetization must have been acquired at least before the Cenozoic.

No paleomagnetic results from the Mesozoic of the Baoshan area have been reported so far. Paleomagnetic data from the Mesozoic of Southeast Asia are still sparse and most of the studies published are preliminary in nature (Haile and Briden, 1982). During the past ten years or more, several paleomagnetic studies were carried out on the Khorat Group of late Triassic to early Cenozoic age in eastern Thailand (Barr et al. 1978; Bunopas, 1981; Maranate, 1982; Achache and Courtillot, 1985). If the Shan–Thai–Malay block collided with the Indochina and South China blocks in the late Triassic as suggested by Metcalfe (1988), the paleomagnetic direction derived from the Woniusi Fm. in this study will be certainly different from those from Triassic through Cretaceous rocks. The directions obtained from these rocks are mostly northeasterly with moderate, positive inclinations distinctly different from those seen in this study. Furthermore, Chen and Courtillot (1989) studied thirteen sites from the Khorat plateau with ages spanning from the Devonian to the Jurassic, and suggested that most of the Indochina block suffered complete remagnetization associated with the collision of India and Eurasia. If this is true, then the Carboniferous results presented here must have survived this widespread remagnetization event because they pass the fold test while others do not.

So it seems that the ChRM directions isolated from the Woniusi Fm. were acquired prior to the Mesozoic. The Late Carboniferous to Late Permian is a period during which the Earth's magnetic field was dominantly reversed in polarity (Irving and Pullaiah, 1976). Although our sampling does not cover the whole basaltic sequence, the fact that only one polarity is present in all the samples from two localities is consistent with the well established Permo-Carboniferous Superchron.

As for the paleomagnetic data from the Paleozoic from the Shan–Thai–Malay block, there are two important data sets presently available. One was derived by McElhinny et al. (1974) from the Lower Carboniferous to Lower Permian clastics and rhyolites from West Malaya; another was reported by Fang et al. (1989) from Devonian limestones from the Baoshan block. Since McElhinny et al's results are of reconnaissance nature and were discussed by Fang et al., we would like to focus our discussion on the interpretation of our Carboniferous results in comparison with their Devonian ones.

Fang et al. sampled the Devonian rocks essentially from the same area as our first locality, with the farthest sites being at most 50 km south of the type section for the Woniusi Fm. From 31 samples they isolated dual-polarity ChRM directions which pass the fold test and yield a mean of $D/I =$ 18.0°/62.0° with $\alpha_{95} = 5.8°$. A Devonian age was assumed for the magnetization. The paleo-inclination is essentially the same as that of the Upper Carboniferous obtained by us from the Baoshan locality, but the paleo-declinations are almost opposed (different by 170°). As a consequence, if the Shan–Thai–Malay block is restored to the southern hemisphere during the Devono-Carboniferous time as discussed below, large amounts of rotation would be required to reconcile the data presented for the Devonian as well as the two localities sampled for the Carboniferous.

There is a general consensus among earth scientists working in Southeast Asia that the Shan–Thai–Malay block originated from the northern margin of Gondwanaland (Ridd, 1971, 1980; Audley-Charles, 1983, 1988; Mitchell, 1981; Stauffer,

1983; Şengör, 1979, 1984, 1987; Metcalfe, 1988). The major arguments for this are: (1) the stratigraphic evidence shows that there was a landmass west of Malay and Thai peninsula in the early Paleozoic (Jones, 1968; Ridd, 1971); (2) a Triassic active margin occurs along the eastern side of the block (Ridd, 1980; Mitchell, 1981; Şengör, 1987); (3) the Early Paleozoic (Cambro-Ordovician) marine faunas show close affinities to comparable faunas of northwest Australia (Burrett and Stait, 1985, 1986), and the Early Permian brachiopods also show affinities with northwest Australia, New Guinea, and Timor (Archbold et al., 1982); (4) Early Permian brachiopods and foraminifers indicate cool- or cold-water conditions on the block (Waterhouse, 1982; Fang, 1983); and (5) there occurs an extensive belt of glacio-marine diamictites of late Carboniferous–early Permian age extending from west Yunnan southwards to Sumatra (Stauffer, 1983; Stauffer and Lee, 1986; Wang, 1983; Cao, 1986). Based on this geologic, biogeographic and paleoclimatologic evidence, a popular late Paleozoic reconstruction involving the Shan–Thai–Malay block has emerged in recent years, in which the elongate Shan–Thai–Malay block, comprising northwest Sumatra, west Malaya, Peninsular Burma and Thailand, northwest Thailand, the Shan States of Burma, west Yunnan of China (Baoshan area), and, in some versions, southern Tibet (Lhasa block), has been placed as a sliver of continent in the northern periphery of Gondwanaland, with its present western side against northwestern Australia and northeastern India (Audley-Charles, 1983; Şengör, 1987; Metcalfe, 1988; Nie et al., 1990). The overall orientation of the Shan–Thai–Malay block in this paleogeographic reconstruction is E–W or ENE–WSW. If this is correct, it would be unavoidable that a rotation of more than 100° has to have taken place since the late Paleozoic in order for the Shan–Thai–Malay terrane as a whole to reach its present NNW–SSE orientation. On the other hand, if the block is reconstructed for the late Paleozoic paleogeographic position according to the paleomagnetic declination determined from the Woniusi Fm. from Baoshan, the present eastern side of the block would have to be attached to northwest Australia instead, and Malay and Sumatra would be at as high latitudes as 60–70°, both conflicting with the existing geologic data.

Thus, the large rotation of the Shan–Thai–Malay block predicted from the Devonian results appears to be concordant with the known data. How then to explain the big difference in paleomagnetic declinations observed from the Devonian and Carboniferous rocks from the adjacent sites?

In comparison with the Mesozoic–Cenozoic section of the apparent polar wander (APW) path for Gondwana, the paleopoles for the Paleozoic section are fewer and with more uncertainties (Schmidt et al., 1990). Some early studies of the 1950's and 1960's should probably better be redone with more advanced techniques and more sophisticated instruments now available. It should not come as a surprise therefore that different versions have been proposed for the Paleozoic APW path for Gondwana (see Van Houten and Hargraves, 1987; Schmidt et al., 1990). Nevertheless, increasing data indicate that the paleomagnetically determined South Pole for Gondwana was located in central Africa in the Late Devonian (Van der Voo, 1988; Kent and Van der Voo, 1990), and that the South Pole then moved southward out of Africa and swung eastward across central Antarctica during the Carboniferous and Permian (McElhinny and Embleton, 1974; Vilas, 1981; Bachtadse and Briden, 1990). As can be seen from Fig. 7, if the Shan–Thai–Malay block was rigidly attached to Eastern Gondwana during the Late Devonian through Early Permian, the paleodeclination for the Baoshan area would have a change of about 70° in response to the northward drift and clockwise rotation of Gondwana. The actual change in paleo-declination during the Late Devonian through Late Carboniferous should be less than this, since the Late Carboniferous pole for Gondwana would be located somewhere between the Early Permian pole and the Early Carboniferous pole. The paleolatitude we observed from the Woniusi Fm. is lower than what would be expected (ca. 58°) if the Late Carboniferous pole is close to the Early Permian pole or if the direction observed was acquired in the Permian.

The Devonian results reported by Fang et al.

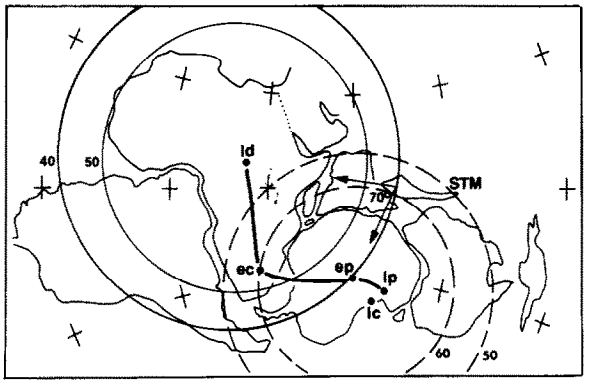


Fig. 7. Late Devonian through Permian APW path for Gondwana, showing the possible change in paleo-declination (arrows) in the Baoshan area of the Shan–Thai–Malay (STM) block due to the apparent polar shift during this time interval. See text for further discussion. Gondwanan reconstruction follows Smith and Hallam (1970), and the paleoposition of the STM block follows Fang et al. (1989). Dots denote the poles used: *ld* = Late Devonian from Canning Basin Limestone, Western Australia (Hurley and Van der Voo, 1987); *ec* = Early Carboniferous from Dwyka glacial varves, Central Africa (McElhinny and Opdyke, 1968); *ep* = Early Permian from Upper Serie d'Abadla redbeds, Morocco (Morel et al., 1981); *lp* = Late Permian from Upper Marine Latites, Eastern Australia (Irving and Parry, 1963); *lc* = Late Carboniferous from Upper Kuttung sediments, Eastern Australia (Irving, 1966). This pole might be affected by Permian overprint as suggested by its proximity to the Permian poles. The large solid (broken) circles are paleolatitudes around the Late Devonian (Early Permian) pole. Equal-angle projection.

involve the Lower, Middle and Upper Devonian rocks. If the paleomagnetic declination they determined could represent that of the whole Shan–Thai–Malay block, a clockwise rotation of more than 130° must have taken place post-Carboniferous before the block reached its present orientation. It is difficult to determine whether this occurred largely when the Shan–Thai–Malay block collided with the Indochina and South China blocks or some time later. To explain the remaining difference in paleo-declination between the Devonian and the Upper Carboniferous results, however, minimum tectonic rotations of more than 100° and 15° counterclockwise have to be required respectively for the Baoshan and Yongde locality relative to the sampling sites for the Devonian. The middle Cretaceous rock formations in the neighboring South Yunnan basin on the Indochina block have been shown to be rotated clockwise by up to 65° with respect to stable Eurasia (Opdyke and Huang, 1990). Our newly obtained, unpublished Middle Jurassic results from a locality 140 km south of Baoshan city indicate a 85° clockwise rotation relative to stable Eurasia. A recent paleomagnetic study by Schmidtke et al. (1990) also demonstrates up to 108° of counterclockwise rotation of Borneo relative to stable Eurasia during the Cretaceous and Cenozoic. All this reveals the extremely fragmental nature of this part of the world; the tectonic rotations are probably mostly, if not all, related to the collision of India with Asia. The kinematic pattern being revealed in Southeast Asia seems even more complex than that of western North America (Beck, 1980).

Conclusion

A re-study of the paleomagnetism of the Upper Carboniferous Woniusi Fm. from Baoshan confirms the previous workers' results, and a positive fold test further constrains the acquisition of the magnetization to be prefolding. The paleomagnetic inclinations determined from two sampling localities for the Woniusi Fm. of this study are similar to that observed from the Devonian rocks from the same area by other workers, implying a 42° paleolatitude for the Baoshan area during the Devono-Carboniferous time. The observed paleomagnetic declinations, however, present large difference between the two localities for the Upper Carboniferous as well as between those for the Upper Carboniferous and the Devonian. It could be partly explained by the apparent polar shift of Gondwana, assuming that the Shan–Thai–Malay block was rigidly attached to Eastern Gondwana during this time interval. A large part of the paleo-declination difference has to be ascribed to within-block rotation between the adjacent sampling sites, which is probably related to the collision of India with Asia. The unraveling of the tectonic evolution of Southeast Asia, which has such complex structure and history, requires abundant paleomagnetic results of high quality and an expansion of sampling to rocks of Cenozoic age.

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